

MANU MARTIN

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Visa: MATES (Subclass 403) – Unrestricted Work Rights, 2 Years

PROFESSIONAL SUMMARY

Software Engineer and AI Specialist with hands-on experience connecting production-grade ML models to Java/Spring Boot and Python backends. Built and shipped reactive REST APIs, computer vision pipelines on edge hardware, and LLM-powered automation that cut manual workload by 70–90%. Comfortable owning full-stack features end-to-end and mentoring junior developers through technical transitions.

PROFESSIONAL EXPERIENCE

Iolite Technologies & Christ University Ecosystem — Bengaluru, India

Associate Software Engineer (Iolite Technologies) – Apr 2026 – Present

- Co-built the EREC Service module using Java, Spring Boot, and Spring WebFlux — replacing a manual paper-based ethics-clearance workflow and cutting average approval time from ~5 days to same-day for most submissions.
- Developed non-blocking reactive REST APIs (WebClient) to handle concurrent research submission spikes without degrading response times.
- Built the Reviewer Pool Setup UI in React.js + Material-UI, giving admins dynamic workload controls that kept SLA breaches at zero across the first two submission windows.
- Integrated a Python generative AI microservice for automatic PDF parsing, questionnaire pre-population, and checklist compliance checks — removing ~3 hours of per-submission manual review.
- Ran pair-programming sessions for two junior developers transitioning to reactive programming and LLM integrations, reducing their ramp-up time by roughly half.

AI Developer (Christ University Contract) – Jan 2026 – Apr 2026

- Automated repetitive ERP data-entry workflows using fine-tuned ML models on custom university datasets, reducing required manual intervention by over 90%.
- Designed and optimised prompt structures for GPT-based LLMs to deliver contextually accurate outputs within the university's compliance domain.
- Led internal tech discussions and pair-programming sessions on LLM alignment and prompt engineering to upskill peers on practical AI tooling.

Freelance AI Technologist & Data Scientist | MR Technologies | Remote | May 2025 – Jan 2026

- Built a vision-based assistive medication-compliance system using OpenCV and ONNX, deployed to a Raspberry Pi; achieved real-time inference at <200 ms per frame for a high-risk patient pilot.
- Designed and optimised image-processing pipelines for edge gesture-classification models, keeping CPU utilisation under 40% on constrained hardware.
- Delivered NLP-focused AI applications for multiple clients, covering data cleaning, feature extraction, and model deployment.

Conversation Designer Intern | Comportement Software Pvt Ltd | Remote | Jan 2025 – May 2025

- Designed dialogue trees and conversational scripts for tier-1 clients Dream11 and ICICI, collaborating directly with senior engineers.
- Ran UAT sessions to identify response-clarity gaps, iterating scripts until error rates dropped to acceptable thresholds.

SELECTED PROJECTS

Research Ethical Clearance Portal (EREC) — Christ University ESPRO

- Full-stack automated clearance engine (Spring WebFlux + React) cutting ethics-approval bottlenecks and generating secure PDF certificates automatically on final validation.

Alphonso — On-Device AI Content Filter — github.com/ManuMartinDeveloper/Alphonso

- Migrated Android content-screening from NudeNet 320n to YOLO v8 architecture, cutting inference latency by ~60% while keeping all processing on-device for privacy.

- Added Isolation Forest anomaly detection for smart lockdown of unusual usage patterns without cloud dependency.

Fr. Francis — Catholic Priest AI Chatbot — github.com/ManuMartinDeveloper/Fr_Francis_Chatbot

- Streamlit assistant built on LangChain + multi-source RAG over the Groq API for sub-second theological query responses across a 50k+ document corpus.

Eva — Automated Job-Seeking Assistant — github.com/ManuMartinDeveloper/Eva-AI-Job-Butler

- Multi-agent pipeline that scans job boards, scores listings against a resume, and logs pipeline state automatically — reducing manual tracking overhead by 70%.

Smart Blind Stick — YOLO + Raspberry Pi —

- Embedded object-detection (YOLOv5) on Raspberry Pi with audio feedback loop; navigates obstacles in real time for visually impaired users.

TECHNICAL SKILLS

- Languages: Python, Java (Spring Boot, Spring WebFlux), JavaScript, SQL, C
- Frameworks & Libraries: React.js, Material-UI, TensorFlow, PyTorch, Scikit-learn, ONNX, Hibernate, FastAPI, Streamlit
- AI / ML: Transformers, LangChain, RAG Pipelines, Prompt Engineering, YOLO, OpenCV, Isolation Forest, Behaviour Modelling
- Infrastructure & Tools: AWS, Firebase, Docker, Git, Power BI, Tableau, Nextcloud

EDUCATION

Master of Science – Artificial Intelligence and Machine Learning

Christ (Deemed to be) University, Bengaluru | 2023 – 2025 | Thesis: Sentiment Analysis with Large Language Models

Bachelor of Science – Computer Science, Mathematics & Electronics

Christ (Deemed to be) University, Bengaluru | 2020 – 2023 | Capstone: IoT FarmChat Project

ADDITIONAL

- Interests: Assistive technology, self-hosted infrastructure, open-source contributions, travel & photography